

NICHOLAS LAM

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EDUCATION

Singapore Management University

Singapore

Bachelor of Science (Information Systems)

Aug 2023 - May 2027 (Expected)

Product Development Track

GPA: 3.74/4

TECHNICAL SKILLS

- Programming Languages: Python, HTML, CSS, Javascript, Typescript, SQL, Java
- Tools & Frameworks: React, Vue.js, Bootstrap, Node.js, Next.js, Tailwind CSS, OpenCV, SpringBoot, Expo
- Databases: MySQL, Supabase, Firebase, Firestore
- Software Development Tools: Figma, Vercel, Netlify, Docker, Postman, Ollama, Azure, AWS, Flask, Anything LLM, Kubernetes, Outsystems
- Design Software: Photoshop, Illustrator, Premiere Pro, Blender

EXPERIENCE

Artistry Heals

Full Stack Developer

Jan 2025 - Present

- Designed and developed a cross-platform React Native (Expo) app (iOS/Android/Web) featuring three interactive micro-games that combine breath, motion, and touch to reduce tension.
- Engineered real-time audio, haptics, sensors, and animations (Expo AV, Haptics, Accelerometer, Reanimated, Skia/SVG, Matter.js) to deliver responsive, 60 FPS immersive experiences.
- Delivered end-to-end ownership across architecture, audio/sensor reliability, data model design, and platform-safe configuration as well as file-based routing with expo-router for robust navigation handling.
- Integrated Supabase with typed APIs to persist onboarding, pre/post-session surveys, and gameplay outcomes, to enable measurable stress-reduction insights.
- Revamped and launched the ArtistryHeals marketing website into a responsive, production-ready web app using React, Tailwind, and Auth, with EmailJS integration for contact forms.

PROJECTS

Dungeons End

Docker Project

Jan 2025 - April 2025

- Text-based dungeon crawler game using Docker containers, adhering to microservice architecture principles. Included inventory management, character stats, and combat mechanics.
- Developed a text-based dungeon crawler using a modular microservice architecture (14+ Python/Flask services) orchestrated via Docker Compose, RabbitMQ, and MySQL.
- Engineered a resilient web gateway and composite service flows enabling login, traversal, combat, inventory, scoring, and session management with robust error handling.
- Implemented asynchronous logging and real-time monitoring, persisting audit trails in MySQL via RabbitMQ while ensuring a responsive retro jQuery/Flask UI with combat logs, health bars, and stat panels.

Passport photo maker

Java Project

Jan 2025 - April 2025

- Built a full-stack web app (Java/Spring Boot, React/Tailwind) that removes photo backgrounds using U2Net (ONNX Runtime) and applies adjustments (brightness/contrast/saturation)
- Engineered a high-quality image pipeline combining deep learning, OpenCV morphological refinement, and GrabCut to achieve crisp alpha mattes and professional compositing across diverse skin tones and clothing.